

DECSELFMASK: Leveraging Unlabeled Text via Self-Relevance-Guided Masking for Decoder-Only Classification

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Abstract

Classification tasks require annotated data, which can often be expensive, time-consuming, or even unfeasible to collect. This is the case of the medical domain, where large datasets often have few annotated examples. To address this, we propose DecSelfMask (Decoder Self-learning by Masking), an approach to enhance decoder-only performance on classification tasks. We build on common self-learning approaches by leveraging a model to create training examples from unlabeled data, and propose a novel relevance-guided masking strategy. We use relevance attribution methods to determine what portions of unannotated texts are relevant for a task. We then create self-supervised training examples by masking out those portions, training the model to reconstruct them via next-token-prediction. We hypothesize that those examples convey knowledge about the structure and semantics of unannotated data that can be useful for downstream performance. We test our approach on 136 tasks from a collection of 1.9M clinical notes from an Italian hospital. We quantify DecSelfMask’s impact on downstream tasks on 5 models of different scales and families, including a probing analysis. Experiments show consistent gains, outperforming the base models (+9.1 points in Macro F1), continual pretraining (+6.3), synthetic label generation (+12.5), as well as common baselines. The results show that relevance attribution can serve as a powerful source of supervision for decoder learning from unlabeled data, beyond existing methods.

1 Introduction

Training automatic systems to perform Natural Language Processing tasks requires large amounts of high-quality annotated data (van Engelen and Hoos, 2020; Yakimovich et al., 2021). Annotations collection can be performed by non-experts (Snow et al., 2008), but there are cases where high degrees of expertise are required (Yang et al., 2019). Such

processes can be expensive and time-consuming, and in domains like medicine they can represent a major issue (Yakimovich et al., 2021).

Because of these challenges, medical applications often involve an abundance of raw data, and limited annotated examples. Common approaches to tackle these scenarios rely on *self-learning* (Wang et al., 2023), where manual annotations are first leveraged to generate synthetic ones, which are then combined with the former to train downstream models (Shi et al., 2023; Luo et al., 2025). Other approaches use raw data to perform *continual pretraining*, aiming to adapt models to the target domain (Singhal et al., 2023; Wu et al., 2024), building on the implicit relation existing between the raw text and the downstream task.

While *self-learning* tries to project such knowledge into unannotated text and expand the task-oriented training set, *continual-pretraining* attempts to gain domain-oriented knowledge from the data itself, without explicitly considering the target tasks. An interesting direction remains underexplored at their intersection, combining the acquisition of knowledge from raw data and the guidance that can be given by annotated examples. For instance, a target task on “*asthma*” classification can provide guidance to automatically identify portions of a raw text such as “*frequent stitches when breathing*” and “*critical respiratory rate*”, to learn the implicit relation existing between them, and potentially increase downstream performance on the target task itself (Figure 1).

Following this hypothesis, we propose **Decoder Self-learning by Masking** (DECSELFMASK), a method to create training examples from the knowledge embedded in unannotated data. At its core, it is a self-learning-based technique to automatically generate masked examples to train decoder-only models, specifically targeting relevant portions of the input. Our contribution is a theoretical framework and the experimental evidence that relevance

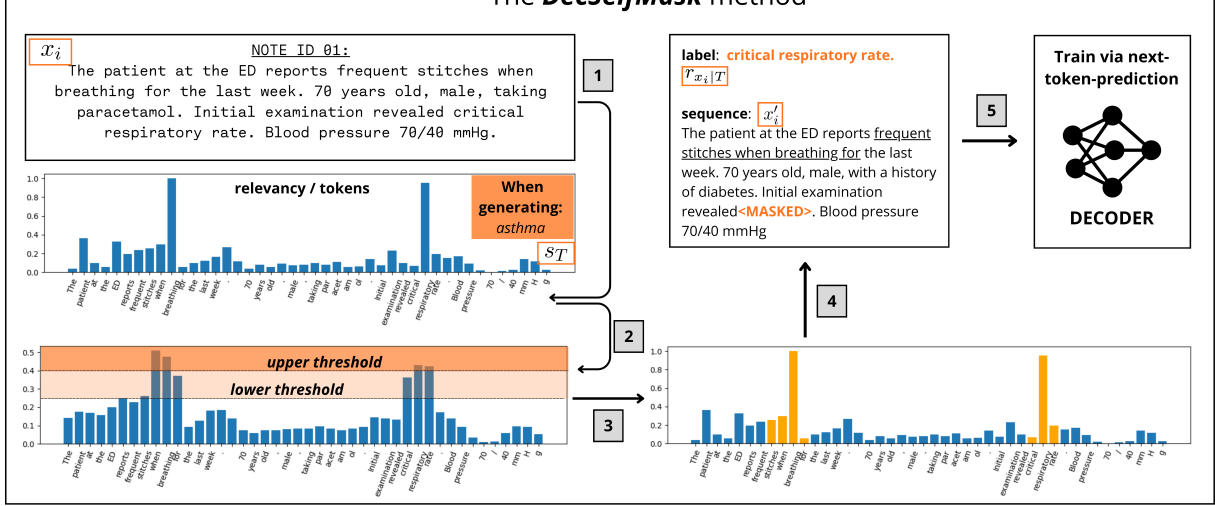


Figure 1: **DECSELF MASK method overview**, exemplified by the task of determining if a patient has asthma. **1) Relevancy calculation:** the raw text x_i is pre-pended to the target text portion s_T (e.g., *asthma*) and given as input to the model. A relevance score is computed for each input token with respect to s_T . **2) Gaussian smoothing:** the relevance scores are smoothed and two thresholds are identified. **3) Groups generation:** portions of relevant text $r_{x_i|T}$ are identified by all consecutive tokens with relevance higher than a lower threshold, where at least one token has relevance higher than the lower threshold. **4) Masking:** training sequences x'_i are generated by masking out relevant portions of text x_i . **5) Training:** decoders are trained via next-token-prediction on the generated examples.

attribution can itself serve as a powerful source of supervision for decoder learning from unlabeled data. It can be articulated as follows:

- We propose DECSELF MASK, a method to combine training-based domain adaptation (continual pretraining) with self-supervised learning, aiming to increase performance on classification tasks.
- We evaluate DECSELF MASK on medical 136 tasks, finding an increase from standard approaches of +19.9 Macro F1 points on average among models scales and families.
- We compare DECSELF MASK to several baselines and state-of-the-art approaches, with gains of +6.3 over continual pretraining and +12.5 over self-learning.

We release the codebase to utilize our framework on custom data and replicate the experiments¹.

2 Related Work

Semi-supervised Learning and Self-training.

Automatic systems can be *supervised* or *unsupervised* (Bishop, 2006). When labeled data is present but scarce, mixed *semi-supervised* approaches have

shown to be effective (Zhu, 2008). With the rise of LLMs, these methods have shifted toward *self-learning* (Singhal et al., 2023), where models are trained on the labeled examples aiming to generate synthetic-labels from unannotated data, to be then utilized for further training (Gera et al., 2022; Shi et al., 2023; Wang et al., 2023; Luo et al., 2025). These methods consist in bootstrapping model’s own knowledge about the downstream task. By contrast, we use models’ own estimates to automatically identify task-relevant information in raw text and transform it into self-supervised training signal, without explicit reference to the task itself.

Medical Domain Adaptation via Continual Pre-training.

With the advent of LLMs, there has been a growing interest in models adaptation to the medical domain. The main technique is continual pretraining (CPT) (Ou et al., 2025), where models are exposed to large corpora of raw medical data via next-token-prediction, as this requires no expert labels. Examples are BioGPT (Luo et al., 2022), MedPALM-1 (Singhal et al., 2023), PMC-LLaMA (Wu et al., 2024), MedPalm-2 (Singhal et al., 2025), MedPhi (Corbeil et al., 2025), MedGemma (Sellersgren et al., 2026). These methods have shown to be effective by aligning models’ knowledge to the target domain, without directly modeling downstream tasks. Notably, oth-

¹<https://github.com/ferrazzi Pietro/DecSelfMask>

ers have shown CPT having low-to-no impact on downstream medical tasks (Ferrazzi et al., 2026a; ?) suggesting that alternative approaches should be explored. We follow this line by introducing task-informed signal to CPT.

Decoder-only Masking Strategies. Dukić and Šnajder (2024); Huang et al. (2025) propose to act on decoders’ causal mask to improve performance on sequence labeling and question answering tasks. More similar to us, Khosla et al. (2025) present a method to enable decoders to generate sentence-level representations to perform text in-filling using bi-directional context, yet not performing any self-learning nor task-oriented optimization. By contrast, we do not modify the models’ structure in any way.

Relevance Attribution in Transformers. Several methods have been proposed to determine the relevance of the input for a given output generated by a transformer-based model (Ferrando et al., 2024). While perturbation approaches (Covert et al., 2021) and contrastive attribution (Yin and Neubig, 2022) require extensive comparisons and alternative search, gradient-based methods can be applied independently from the context. Hence, we consider AttnLRP (Achtibat et al., 2024) for input relevance attribution in the context of our method.

3 Problem Formulation and Hypothesis

In this work, we propose a method to leverage the structure of unlabeled, domain-specific corpora to solve downstream classification tasks. Our approach relies on the hypothesis that if relevant information can be automatically identified, it can be exploited to construct self-supervised learning objectives, aligned with the downstream task.

Learning from Unlabeled Data. Let us define $X = \{x_i\}_{i=1}^N$ denoting a collection of sequences sampled from a domain-specific corpus, and let $C = \{c_i \in \{c_1, \dots, c_K\}\}_{i=1}^N$ denote the relative class labels for the classification task T , which are unknown. Following van Engelen and Hoos (2020), we investigate whether the marginal distribution over the input space $P(x)$ contains useful information for learning the posterior distribution $P(c | x)$. In our setting, a decoder-only language model $\text{LLM}(c | x)$ is used to estimate this conditional distribution. Our central hypothesis, similar to the one behind continual pretraining, is that the

conditional distribution $P(c | x)$ is not independent from the structure of the input distribution $P(x)$. Consequently, we hypothesize that learning regularities, dependencies, and semantic structures within the unlabeled examples sampled from $P(x)$ can improve performance on the downstream task $T \sim P(c | x)$. However, the relationship between $P(x)$ and $P(c | x)$ is generally implicit and task-dependent (Oliver et al., 2018). In what follows, we describe our approach to verify this hypothesis.

Linking $P(c|x)$ and $P(x)$ through LLM. Classification tasks rely on the fact that given an input sequence x_i and a task T , there are portions of x_i that are relevant to predict the task-related label c_i . Let

$$r_{x_i|T}(\text{start}, \text{end}) = \{x_{i,j}\}_{j=\text{start}}^{\text{end}} \subset x_i \quad (1)$$

denote a subset of tokens in x_i identified as *relevant*² for the downstream task T . Instead of directly learning $\text{LLM}(c_i | x_i)$, we define an auxiliary self-supervised objective:

$$\text{LLM}(r_{x_i|T} | x'_i), \quad (2)$$

where the model predicts a relevant portion of the input $r_{x_i|T}$ from $x'_i = x_i \setminus r_{x_i|T}$, the input text itself where the relevant portion has been masked out (Figure 1, top right). This formulation resembles self-learning, where the model itself identifies the labels to learn from. Our hypothesis is then framed as follows: the LLM can be trained on the objective defined in Equation 2 (which solely depends on the input $P(x)$) to better approximate $P(c|x)$.

In other words, we hypothesize that learning to auto-regressively predict task-relevant portions $r_{x_i|T}$ from x'_i encourages the model to acquire domain-specific knowledge, linguistic structures, and latent correlations that are also beneficial for the downstream task T .

The "Two Peaks" Intuition. We aim to train models on those inputs x_i that convey some relevant knowledge $r_{x_i|T}$ about the downstream task T , according to the structure of Equation 2. On the other hand, we rely on utilizing $r_{x_i|T}$ as a label that needs to be predicted, masking it out from the training sequence x'_i . Then, there is no guarantee that, after masking, x'_i will convey enough task-relevant information to properly predict $r_{x_i|T}$ in the context of T . Therefore, we include in our training set only

²For simplicity, in the following we refer to $r_{x_i|T}(\text{start}, \text{end})$ as $r_{x_i|T}$

sequences with two or more disjointed relevant portions $r_{x_i|T}(s_1, e_1)$ and $r_{x_i|T}(s_2, e_2)$ (Figure 1, bottom right). By doing so, we leverage the self-learning framework to its full extent, by training models to predict one portion of text that the model itself identified as relevant to the task, ensuring at least another one exists to provide enough context for this to be possible. We refer to this idea as the "two peaks" framework, given that we only select sequences with at least two peaks of relevance.

Relevance Estimation. Our method relies on the ability to extract portions $r_{x|T}$ that are somehow *relevant* for the downstream task T , to then expose models to that information. Following the concept of *self-learning*, we compute such relevance by leveraging the model itself. We append to each input x_i a short string s_T that describes the task T (in the example of Figure 1, $s_T = \text{"asthma"}$). Then, we use AttnLRP (Achtibat et al., 2024) to quantify the relevance attributed by the model itself to each token in x_i if it was generating s_T . For a given sequence of input tokens $x_i = (x_{i,1}, \dots, x_{i,n})$, AttnLRP provides a relevance score for each token $\text{rel}_{x_i|s_T} = (\text{rel}_{x_{i,1}|s_T}, \dots, \text{rel}_{x_{i,n}|s_T})$.

Training Data Selection. We define a sequence x_i as relevant for a task T if there exist at least two disjoint subsequences where all tokens have relevance higher than a lower threshold t_l , and at least one token has relevance higher than an upper threshold t_u (following the "two peaks" framework). Formally, each of the two disjoint subsequences $r_{x_i|T}(s, e)$ complies with the following conditions:

$$\begin{aligned} \text{cond}_1(s, e) : G(\text{rel}_{x_{i,j}|s_T}) &\geq t_l \quad \forall j \in [s, e] \\ \text{cond}_2(s, e) : \exists j \in [s, e] \text{ s.t. } G(\text{rel}_{x_{i,j}|s_T}) &\geq t_u, \end{aligned} \quad (3)$$

where $G(\cdot)$ is a gaussian smoothing function on the relevance scores (Figure 1, step 2).

Since $\text{rel}_{x_{i,j}|s_T}$ is defined over tokens, $r_{x_i|T}$ spans may not align with full words and can split sub-word tokens. For smoothing purposes, we extend each span to cover the complete words intersecting the selected token range.

The objective proposed in Equation 2, combined with the training data defined in this manner, acts as a form of task-oriented self-supervision: the model extracts supervision signals directly from unlabeled data by leveraging the fact that it has identified as relevant for the task at least another portion of text.

4 Experimental Setup

To verify the hypothesis defined in Section 3, we require a large dataset of **unlabeled data** together with a smaller set of **expert-annotated data** for medical classification tasks. We leverage the SGB dataset of medical text presented by Ferrazzi et al. (2026b), released with a permissive license. This comprises around 1.9 million anonymized clinical notes from the Emergency Department of the San Giovanni Bosco Hospital (Turin, Italy).

Ferrazzi et al. (2026c) further proposed the Case Report Form (CRF) filling task built on top of the SGB dataset. The task consists of filling a list of 136 medical items with the correct values, given a clinical note as input. We reshape the task to make it resemble pure **classification** and obtain a set of 136 individual classification tasks, one for each of the original CRF items. Each classification task is characterized by three or more possible labels. For instance, the classification task defined by the string s_T "heart rate" consists in determining whether a patient is "bradycardic", "normocardic", "tachycardic", or if there is not enough information to determine so ("unknown"). We extended the dataset to 6404 notes³, each annotated for one or more relevant tasks, resulting in 61k pairs of note-annotation in total. Each task has an average of 440 annotated examples (variance of 147). Table 5 (Appendix) reports all tasks and their label space.

Furthermore, we define two **held-out classification tasks** on notes from the SGB dataset, determining if a patient presents any "chronic" condition, or determining whether a patient is entering the emergency department with "dyspnea" or "loss of consciousness (tloc)". We reserve the two tasks for evaluating whether the knowledge acquired during DECSELFmask training can transfer to previously unseen tasks. We believe this setting to be particularly relevant in real-world medical environments, where new classification needs may emerge over time. All datasets are described in Table 1.

Relevancy Calculation. We follow the procedure described in Section 3 to generate the masked training data (Figure 1). For each task T (out of the 136), we need to define the target s_T to base the relevance generation on. In the datasets we use, each T comes with a short text (one to five words) that defines the dimension to classify the note on (e.g. *heart rate*, Table 5 for full list). Therefore, we

³Can be obtained via email to the corresponding author.

Dataset	Description	Use	n	words	n tasks
SGB	Anonymized raw clinical notes from the emergency department of an Italian hospital.	Generation of the DECSELF-MASK training examples.	1.9M	125.7M	—
CRF	Annotated clinical notes from the SGB dataset on 136 medical items. Each note is annotated once for each relevant item.	Guidance on how to construct the DECSELF-MASK training examples; main evaluation.	61k	4.7M	136
Chronicity	Annotated clinical notes from the SGB dataset determining whether a patient has chronic conditions.	Evaluate DECSELF-MASK on tasks not involved in its definition and training.	2587	335k	1
T-D	Annotated clinical notes from the SGB dataset determining whether a patient is admitted to the emergency department with loss of consciousness or dyspnea.	Evaluate DECSELF-MASK on tasks not involved in its definition and training.	1713	331k	1

Table 1: **Datasets used in our experiments**, with number of examples n , *words*, and *tasks* (where applicable). *SGB* is used for DECSELF-MASK learning, to improve performance on classifying notes on the 136 *CRF* items. *Chronicity* and *T-D* are not directly involved in the DECSELF-MASK process, but left for further testing.

can use it as s_T , making sure the relevance calculation is based on text that is relevant for T itself. For each task, we append s_T to each clinical note in the SGB dataset (template is provided in Appendix C).

The calculation of the relevance via AttnLRP must be performed over one single generated token; however, all 136 s_T texts are multi-token sequences. While we could average the relevance calculated for each generated token, this would result in repeated runs over the input. Instead, we use the relevance computed on the middle token of s_T as a computationally efficient proxy. See Appendix A for the validation of this choice. Furthermore, for each note, 136 relevance-calculation prompts can be generated, one per task (defined by s_T). Each of these prompts can result in as many training sequences as identified relevance peaks (minus one). To avoid the explosion of training sequences built on the same note, we randomly sample 9 out of the 136 possible prompts (15%). Overall, we run relevance calculation on one million prompts. This step takes 30 GPU hours on one NVIDIA L40S.

Training Sequences Generation. Once the relevance is obtained for all notes, we perform Gaussian smoothing (kernel size=3, sigma=1), and set an upper (0.4) and lower (0.2) thresholds to identify the groups of relevant tokens. These hyperparameter are selected via manual validation on an held-out set of 100 notes. We set 4 pairs of lower-upper limit (0.2/0.4, 0.2/0.6, 0.4/0.6, 0.5/0.6), and two values for the kernel size of the gaussian smoothing (1, 2). We generate the peaks for each hyperparameter configuration, and count the number of notes where semantically reasonable peaks

have been generated (i.e., peaks that are relevant for the target s_T), and we select the hyperparameter set that provides the best performance. Then, we drop all sequences that do not contain at least two peaks, and perform de-duplication, as different s_T texts might have identified the same token groups as relevant for different tasks T , which would result in identical training sequences.

For each note containing two relevant spans, we generate training examples by masking one span at a time (Figure 1, top right). Due to the large number of generated examples, we further filter the dataset by retaining only sequences in which another relevant span occurs to the left of the masked region. This has two reasons: (i) two relevant sequences from the same clinical note most likely contain similar information, and (ii) encouraging the model to infer the masked content from preceding contextual information, thus mimicking the autoregressive nature of the downstream task.

We collect all the sequences found to be useful for all 136 tasks in one single dataset of 1.03M examples. Each example is composed by a clinical note where a portion has been masked out (to be used as *input*), the text s_T used for the masking process, and the text in the masked portion (to be used as *label*).

DECSELF-MASK Training. To verify if the generated training pairs of input and label can be utilized as sources of knowledge about the downstream classification tasks they were designed for, we analyze three model families in their instructed versions: Llama3, Qwen3 in their ~1B and 8B versions, and Gemma3-4B to represent a mid size be-

Model	SFT			Probing			Δ_{best}
	Base	DSM	Δ_{DSM}	Base	DSM	Δ_{DSM}	
Llama3-1B	21.9 $\pm$ 1.0	19.69 $\pm$ 1.0	-2.2	34.1 $\pm$ 1.1	47.0 $\pm$ 1.3	+12.9	+25.2
Qwen3-1.7B	30.9 $\pm$ 1.2	22.3 $\pm$ 1.0	-8.6	33.3 $\pm$ 1.2	46.8 $\pm$ 1.3	+13.5	+15.9
Gemma3-4B	37.5 $\pm$ 1.2	44.4 $\pm$ 1.3	+7.0	36.1 $\pm$ 1.2	47.5 $\pm$ 1.3	+11.4	+10.0
Llama3-8B	35.3 $\pm$ 1.0	42.9 $\pm$ 1.2	+7.6	55.6 $\pm$ 1.3	55.8 $\pm$ 1.3	+0.2	+20.5
Qwen3-8B	32.5 $\pm$ 1.2	40.4 $\pm$ 1.2	+7.9	52.5 $\pm$ 1.3	60.3 $\pm$ 1.2	+7.7	+27.8

Table 2: **Impact of DECSELF MASK training across task-adaptation strategies.** The *SFT* columns report performance for supervised fine-tuning of the base or DSM models, while the *Probing* columns report performance when using a classification head on the final layer. For each strategy, Δ_{DSM} is the difference between the DSM model and its corresponding base model. The final column reports the performance gain of the best DSM adaptation strategy (either SFT or probing) over the standard SFT baseline applied to the base model. Results are reported as Macro F1 with 95% confidence intervals.

tween the previous ones. We train all parameters for one epoch using the hyperparameters described in Appendix B on two NVIDIA H200 GPUs (requiring a total of 68 GPU hours). The learning objective is next token prediction over the prompt structure presented in Appendix C. By doing so, we obtain 5 DECSELF MASK-models, which can be used to analyze the effect of our method.

5 Evaluation and Discussion

Downstream Tasks Training. We compare the base models with their DECSELF MASK versions, to determine the impact of DECSELF MASK training. We adapt both versions to the downstream tasks via supervised fine-tuning (SFT) on the 136 downstream tasks, and determine which performs better. We divide the CRF dataset into an 85-15 train-test split on an note-item level. For both settings, we train one single multi-task model for all tasks at once, using LoRA (Hu et al., 2022), for one epoch, with the hyperparameters described in Appendix B, for a total of 30 GPU hours.

We calculate performance on all 136 tasks. As aggregation metric to summarize the results on all tasks in one, we use macro F1 following previous work (Ferrazzi et al., 2026c). We compare the performance of base models which undergo task-oriented SFT to our DECSELF MASK-models that undergo the same fine-tuning. As shown in Table 2 (SFT columns), we observe that DECSELF MASK is detrimental for small models ($\sim$ 1B), with an average decrease in performance of -5.4 . On the other hand, it resulted in a homogeneous increase for bigger models, with a $+7.5$ positive impact.

We hypothesize that this behavior is not due to small models failing to acquire useful knowl-

edge with DECSELF MASK, but rather to their limited capacity to coherently generate tokens that reflect this knowledge when prompted to generate the output. In other words, smaller models may successfully encode information that is beneficial for the downstream task, yet lack the representational capacity required to effectively transfer or exploit this knowledge after autoregressive SFT. To verify this hypothesis, further analysis on DecSelfMask-Llama3-1B and DecSelfMask-Qwen3-1.7B is required.

Probing. To better understand why training with DECSELF MASK improves downstream performance only for bigger models, we perform a probing analysis by training a shallow classifier on top of the last hidden layer of the small models ($\sim$ 1B). We add a classification head on the last hidden representation, composed by a linear layer (output size of 256), a ReLU activation with 0.1 dropout, and an output layer to map the 256 neurons to the output classes. We train this head with the hyperparameters described in Appendix B, for an overall of 24 L40S GPU hours. The results are reported in Table 2 (last four columns). We observe a consistent positive effect, suggesting that DECSELF MASK successfully injects useful information into the learned representations, even in cases where this does not translate into improved downstream autoregressive generation performance. The models that underwent our training result in a homogeneous average increase of $+13.2$ points.

From SFT to Probing. The probing analysis suggests that even smaller models learn a useful representation for the downstream tasks. Furthermore, by comparing the results obtained via SFT and

Model	Tloc-Dyspnea			Chronicity			Average Gain	
	SFT Base	CH Base	CH DSM	SFT Base	CH Base	CH DSM	Δ_{CH}	Δ_{SFT}
Llama3-1B	50.5	79.5	84.0	65.8	65.8	69.5	+4.1	+18.6
Qwen3-1.7B	36.9	78.1	86.7	68.9	65.3	68.3	+5.8	+24.6
Gemma3-4B	50.9	85.5	82.6	63.5	66.3	70.1	+0.4	+19.2
Llama3-8B	71.6	80.2	81.3	64.9	64.5	68.1	+2.3	+6.5
Qwen3-8B	47.2	79.3	80.3	56.9	64.7	68.8	+2.6	+22.5

Table 3: **Transfer to Held Out Classification Tasks.** We evaluate whether representations learned through DECSELFmask training transfer to classification tasks not used when constructing the DECSELFmask training sequences. For each task, we report F1 scores for supervised fine-tuning of the base model (*SFT Base*), a classification head trained on the base model (*CH Base*), and a classification head trained on the corresponding DECSELFmask model (*CH DSM*). Δ_{CH} is the average F1 gain of CH DSM over CH Base across the two tasks. Δ_{SFT} is the average F1 gain of CH DSM over SFT Base across the two tasks.

probing, we observed that the potential of the models on these tasks goes much beyond what they exhibit after standard, auto-regressive SFT (an effect that has been analyzed by [Lyu et al. \(2026\)](#)). In fact, Llama3-1B with a classification head before SFT achieves a Macro F1 +11.2 points higher than after standard SFT and without the head (+2.4 for Qwen3-1.7B). Building on this evidence, we train a classification head on top of the larger models as well, to verify if they could achieve better performance. We use the same structure and training parameters described above, obtaining the results shown in Table 2.

We highlight two aspects: first, when adapting models to downstream tasks by adding a classification head, DECSELFmask gives an average improvement of +9.1 in Macro F1, showcasing that the training procedure effectively injects useful knowledge into the models’ representations from the unannotated data. Second, the average performance gain moving from auto-regressive SFT on the base model to classification heads on the DECSELFmask version is +19.9 points, showing this approach to be the most effective.

Held Out Tasks. We reserve two tasks to evaluate whether the knowledge acquired during DECSELFmask training on the 136 medical classification tasks can transfer to new classification settings. We employ two datasets *Chronicity* and *T-D* described in Section 4 and in Table 1. The evaluation metric for both tasks is the F1 score.

We build on the finding that classification heads overperform SFT, and follow this to adapt base models and their DECSELFmask counterparts. In addition, we provide for comparison standard SFT results on the base models, and report all findings in

Table 3. The probing strategy showcases that DECSELFmask results in higher F1 (+3.0 on average across all models) when compared to the base, and it presents an even higher increase of +18.3 when compared with the base with SFT.

Baselines. We compare DECSELFmask to other methods, to contextualize its efficacy. The natural baseline for text classification tasks are **encoder-only models**. We fine-tune on the 136 CRF downstream tasks BERT and its multilingual version multilingual-BERT ([Devlin et al., 2019](#)), BioClinicalBERT ([Alsentzer et al., 2019](#)), ModernBERT ([Warner et al., 2025](#)), and a version of BERT explicitly trained for Italian ([Schweter, 2020](#)). The models are trained to receive as input a clinical note and the text describing the task s_T (e.g., "heart rate") with the hyperparameters defined in Appendix B.

Moreover, we provide the performance of the decoder model that showed the best results in our experiments (Qwen3-8B), prompted **0- and 2-shot**. For 0-shot, we run inference 3 times with slightly different system prompts in order to account for instabilities, following [Sclar et al. \(2024\)](#), and report the average. For 2-shot, we randomly sample the examples, and run inference 3 times to mitigate bias due to their choice. Results are presented in Table 4, showing preference for encoders (up to +13.4 points), which are still outperformed by DECSELFmask.

State of the Art Comparison. Our approach relies on training on the unannotated, raw data. As described in Section 2, the typical approach to expose models to large amounts of raw data is **continual pretraining** (CPT), whose underlying hypoth-

esis inspires DECSELFMASK. We then perform CPT on a model on the same (unmasked) raw clinical notes included in our method. We train using the same hyperparameters as in DECSELFMASK, with the standard next-token-prediction objective. While CPT has a positive impact, it still falls short w.r.t. DECSELFMASK (−6.3 points).

Finally, we include a standard **self-learning** approach, which we take inspiration from to design the training objective of Equation 2. We train a model on the downstream tasks via SFT on the annotated CRF data (similar to the SFT Base models from Tables 2 and 3), and then use it to generate synthetic labels for the unannotated SGB dataset. Then, we train the original base model on the combination of the original and the synthetic labeled data via SFT. While outperforming the base model adapted to the task via SFT (+15.3 points), it still falls behind our approach (−12.5).

Overall, these results highlight that our approach obtains better performance than previous methods, by effectively integrating knowledge embedded in raw data and shaping it through a procedure that enhances classification performance. Moreover, they disentangle the sources of DECSELFMASK’s improvements. While exposure to in-domain clinical data is itself beneficial (as shown by CPT), DECSELFMASK reveals an additional gain attributable to the task-oriented representations learnt by our method. Together with the benefits of exposure to task-related examples, these results suggest that the observed improvements arise not simply from additional data, but from how unlabeled data are transformed into task-relevant training signals.

Two peaks and relevance attribution ablation.

The method we present relies on the "two peaks" criterion (Section 3), meaning that DECSELFMASK models are trained on sequences that include at least two distinct groups of tokens which contain information relevant to the downstream task. To quantify the impact of this theoretical choice, we ablate on the number of peaks, training the best performing model (Qwen3-8B) on sequences exhibiting one single peak (i.e., one single group of relevant tokens). It must be noted that notes with only one peak are less than the ones with more. To account for it, we train on five epochs, reaching comparable training size. We observed that limiting the training to such sequences leads to worse results, with a drop in Macro F1 of −13.1 (Qwen3-8B + one peak + CH in Table 4), experi-

Model	Macro F1
ClinicalBERT	34.1 ± 1.2
multilingual-BERT	34.2 ± 1.2
BERT	36.8 ± 1.2
BERT-Italian	38.2 ± 1.3
ModernBERT	44.1 ± 1.3
Qwen3-8B + 0-shot	27.7 ± 0.6
Qwen3-8B + 2-shot	30.7 ± 0.7
Qwen3-8B + SFT	32.5 ± 1.1
Qwen3-8B + CH	52.5 ± 1.1
Qwen3-8B + CPT + SFT	33.5 ± 1.3
Qwen3-8B + CPT + CH	54.0 ± 1.4
Qwen3-8B + synthetic SFT	47.8 ± 1.3
Qwen3-8B + random mask + SFT	30.4 ± 1.2
Qwen3-8B + random mask + CH	56.4 ± 1.1
Qwen3-8B + one peak + SFT	30.0 ± 1.2
Qwen3-8B + one peak + CH	47.2 ± 1.3
DECSELFMASK-Qwen3-8B-CH	60.3 ± 1.2

Table 4: **Comparison to baselines and state-of-the-art** for the 136 CRF tasks compared with our best performing model (DECSELFMASK-Qwen3-8B-CH with a classification head CH). We provide results for 0- and 2-shot; continual pretraining (*CPT*); self-learning where the model is trained via SFT on synthetic labels generated by itself (*synthetic SFT*). We also showcase the performance of applying DECSELFMASK with *random masking* instead of relying on relevance attribution, and of ablating on the "two peaks" criterion (*one peak*).

mentally motivating the "two peaks" choice.

In addition, we provide results obtained by ablating the relevance attribution step, where we train the model to autoregressively reconstruct the sequences that respected the two peaks criterion, but using randomly masked portion of text instead of the ones found via AttnLRP. Table 4 (Qwen3-8B + random mask) shows that the removal of this component has a negative impact (−3.9 in Macro F1). This ablation highlights that the effectiveness of the signal embedded in the auto-regressive sequence reconstruction depends on the quality of the relevance attribution method.

DECSELFMASK further analysis. Our training procedure modifies the model’s parameters to encode information extracted from unlabeled data, effectively steering the model’s internal representations toward task-relevant patterns present in the raw corpus. To better understand the effect of this training procedure, we analyze how it alters the rel-

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Figure 2: **Impact of the DECSELFmask training on the relevance attributed by AttnLRP to the input for the task "Presence of pacemaker Pacemaker"**. Red represents higher relevance being assign to a specific token by the base model when compared to its DECSELFmask version, while green the opposite. While the base model primarily focuses on terms related to *dyspnea*, the DECSELFmask model shifts its attention toward mentions of "PM" (pacemaker), which is substantially more relevant to the target condition

evance assigned to different portions of the input. More specifically, we provide the model with the same prompt used during the relevance extraction phase of our pipeline, and compute the relevance attributed to the input tokens while generating the target item using the same AttnLRP score used when constructing the dataset.

This analysis reveals notable qualitative differences between the base model and the corresponding DECSELFmask variant. In several cases, our models assign higher relevance to portions of text that are semantically more aligned with the target concept (e.g, Figures 2 and 4 -7 in Appendix).

We quantify this effect by leveraging the labeled clinical notes, where annotators also identified the portions of text considered relevant for each of the 136 tasks. We measure the change in relevance assigned by AttnLRP to these ground-truth spans when moving from the base model to the corresponding DECSELFmask variant. For Qwen-8B, we observe an average relevance increase of 37.2% over the annotated spans, suggesting that the proposed self-supervised training objective effectively steers the model towards assigning higher importance to actually relevant portions of the input.

6 Conclusion

In this work, we introduce DECSELFmask, a task-oriented self-supervised framework to leverage unlabeled, domain-specific corpora to improve the downstream classification performance of decoder-only models, combining the assumptions underlying *self-learning* and *continual pretraining*.

First, we propose to use input attribution methods to mask out task-relevant portions of unlabeled text, and train to autoregressively reconstruct them,

encouraging models to learn domain-specific patterns from raw data, guided by relations identified as relevant to the target tasks by models themselves.

Second, we evaluate DECSELFmask on 136 tasks derived from 1.9M clinical notes. We adapt models to the tasks with a classification head, as probing experiments show consistent improvement over standard SFT across model families and scales (+19.9 F1 Macro points), also showing similar effectiveness on two held-out tasks ad well.

Third, we compare DECSELFmask to standard *continual pretraining* (+6.3) and *self-learning* approaches (+12.5), and propose an ablation study on two components of the method to further highlight the soundness and effectiveness of our approach.

Overall, DECSELFmask contributes a novel way of exploiting unlabeled data for decoder-only classification, by combining ideas from masked training, continual pretraining, and self-learning. In contrast to conventional continual pretraining, it performs task-guided adaptation from unlabeled data, while, unlike existing decoder-only masking approaches, it preserves the standard autoregressive objective and the model’s generative capabilities. Rather than relying on synthetic labels, DECSELFmask uses the model’s own relevance estimates to identify task-relevant spans and turns them into self-supervised training examples. In this way, it exploits the decoder’s prior knowledge of the relationship between the domain and the downstream task, providing a targeted mechanism for learning task-relevant dependencies from unlabeled text. The ablation results further support the central role of relevance-guided masking, showing that learning from the dependencies the model itself considers relevant is particularly effective to acquire task-specific knowledge.

Limitations

Our experiments are limited to Italian clinical notes and relatively small decoder-only models, although results suggest that larger models may benefit more from DECSelfMask. Due to the computational cost of AttnLRP, we train on a subset of the original 1.9M-note corpus and perform joint self-supervised training across all tasks rather than task-specific training. Additionally, the proposed framework assumes that downstream tasks can be represented through short textual descriptions s_T . Future work should explore more efficient relevance extraction methods, alternative masking objectives, and extensions to broader task settings and larger models.

Ethical considerations

Although our approach aims to reduce the dependency on expensive manual annotation, models trained on medical data may still inherit biases, incompleteness, or inaccuracies present in the underlying clinical records. Consequently, the proposed models should not be considered a substitute for professional medical judgment.

Additionally, relevance attribution methods may produce imperfect or misleading explanations of model behavior. Therefore, the relevance analyses presented in this work should be interpreted as approximations of model reasoning rather than faithful causal explanations.

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A Relevance Computation Details

Encoder-only Masking Strategies. The method we present is inspired by language masking training strategies used for encoder-only models. It has been shown that the choice of what portions of text to mask is important when training encoders using masked language modeling (Gu et al., 2020; Ye et al., 2021; Sadeq et al., 2022; Golchin et al., 2023). We take inspiration from this paradigm, shifting it to autoregressive generative models.

Middle Token and AttnLRP Choices. As the cost of calculating the relevance using AttnLRP for all tokens in the output increases linearly with the length of the output, we approximate the average relevance if the input on a certain output using one single output token. A practical challenge arises when the target label is composed of multiple tokens: relevance can be computed using the hidden states associated with different decoding steps, potentially leading to different attribution patterns. In particular, it is unclear whether relevance should be estimated using the first generated token, the last one, or by aggregating information across all label tokens.

To investigate this aspect, we conduct a validation experiment on a medical Named Entity Recognition (NER) dataset (Magnini et al., 2020). For

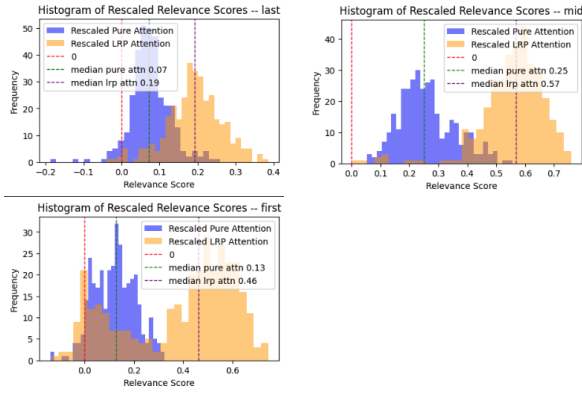


Figure 3: **Comparison between the relevance attributed to ground truth portions of text i) by *Pure Attention* (i.e., average attention across all layers), and *AttnLRP*, ii) when utilizing different tokens of s_T for the relevance calculation.** The x axis represents the increase in relevance that each method is assigning to the ground truth textual portion with respect with uniform distribution (i.e, the more shifted to the right, the better). The best performing configuration is using the *middle* token with *AttnLRP* (top-right histogram).

each annotated entity, we construct a prompt of the form:

You are an entity extractor.
{text} {label}

and evaluate whether the relevance assigned by the model to the tokens in {text} corresponding to the target entity is significantly higher than what would be expected under a uniform relevance distribution.

In parallel, we compare two relevance estimation methods, the average of the attention scores among all layers, and *AttnLRP*, by computing token relevance using the hidden states associated with: (i) the first token of the label, (ii) the last token of the label, and (iii) the middle token of the label. Our results show that using the middle token consistently produces the strongest localization signal, yielding higher relevance scores over the entity span (Figure 3). This suggests that intermediate decoding steps provide a more stable representation of the semantic content of multi-token labels compared to boundary tokens. By doing so, we also validate the usage of *AttnLRP*, which shows the best performance in terms of relevance attribution to the ground truth in the original text.

B Training parameters

DECSELFMASK training We train using deepspeed Zero2 and flash attention, on an effective batch size of 128, using AdamW optimizer with a learning rate of $4 \cdot 10^{-5}$ (cosine scheduler), weight decay of 0.001, warmup ratio 0.1.

Supervised fine-tuning For Supervised fine-tuning, we use a batch size of 64, learning rate of $3 \cdot 10^{-4}$, warmup ratio 0.1, using AdamW optimizer with cosine scheduler and epsilon $1 \cdot 10^{-7}$. We use LoRA (Hu et al., 2022), with r 32, alpha 64, and dropout 0.05. We construct SFT sequences according to the template in Appendix C.

Probing We train for 20 epochs, with learning rate $1 \cdot 10^{-3}$ and the AdamW optimizer.

Encoder-only baseline Training of encoder-only baselines is performed for 10 epochs, with an AdamW optimizer, and a learning rate of $2 \cdot 10^{-5}$. Initial experiments showed that increasing the number of epochs would not provide any benefit.

C Prompts

Relevance calculation We compute the relevance by using the following prompt template:

{clinical note}
Given the patient’s medical history, the following is observed { s_T }.

Note that models’ special tokens are added where required –as well as the added text – but they are excluded from the relevance calculation, following the findings of Xiao et al. (2024).

DECSELFMASK training The self-training is performed using the following template:

Fill in the masked word in the following sentence.
{clinical note} {label}

Note that the simple system prompt is provided to leverage models instruction following, and models’ special tokens are added where required to determine the beginning and end of the instruction and output.

Supervised fine-tuning When fine-tuning DECSELFMASK models on the downstream CRF classification tasks (each task identified by a short description s_T), we combine the data examples according to the following template:

Fill in the masked word in the following sentence.
{masked clinical note}
<crf_item> {s_T}? {masking token} <\crf_item>
<options>{options}<\options>
{label}

The masking token is a model-dependent special token that is used to substitute the masked text.

On the other hand, when fine-tuning base models (e.g., Qwen/Qwen3-8B) on the same tasks, we initially kept a different template, removing the masking token. Then, we observed that changing the prompt did not impact the performance, and kept the previous versions for all. This finding is consistent with [Lyu et al. \(2026\)](#).

options	items
A; V; P; unknown	['level of consciousness']
bradycardic; normocardic; tachycardic; unknown	['heart rate']
bradypneic; eupneic; tachypneic; unknown	['respiratory rate']
certainly active; possibly active; certainly not active; unknown	['active neoplasia']
certainly chronic; possibly chronic; certainly not chronic; unknown	['chronic pulmonary disease', 'chronic respiratory failure', 'chronic cardiac failure', 'chronic renal failure', 'chronic metabolic failure', 'chronic rheumatologic disease', 'chronic dialysis']
current; past; unknown	['presence of respiratory distress']
hypotensive; normotensive; hypertensive; unknown	['blood pressure']
hypothermic; normothermic; hyperthermic; unknown	['body temperature']
measured; unknown	['spo2', 'ph', 'paO2', 'pacO2', 'hcO3-', 'lactates', 'hemoglobin', 'platelets', 'leukocytes', 'c-reactive protein', 'blood sodium', 'blood potassium', 'blood glucose', 'creatinine', 'transaminases', 'inr', 'troponin', 'bnp or nt-pro-bnp', 'd-dimer', 'blood calcium', 'serum creatinine kinase', 'blood alcohol', 'blood drug dosage', 'urine drug test']
pos; neg; unknown	['carotid sinus massage', 'supine-to-standing systolic blood pressure test', 'blood in the stool', 'sars-cov-2 swab test']
short; long; unknown	["duration of the patient's consciousness recovery", "duration of the patient's unconsciousness"]
walking independently; walking with auxiliary aids; walking with physical assistance; bedridden; unknown	['level of autonomy (mobility)']
y; n; unknown	['first episod of epilepsy', 'known history of epilepsy', 'history of allergy', 'history of recent trauma', 'pregnancy', 'history of drug abuse', 'history of alcohol abuse', 'anticoagulants or antiplatelet drug therapy', 'presence of prodromal symptoms', 'compliance with antiepileptic therapy', 'tloc during effort', 'tloc while supine', 'antiepileptic therapy already in place', 'drowsiness, confusion, disorientation as postcritical state', 'stiffness during the episode', 'drooling during the episode', 'tonic-clonic seizures', 'poly-pharmacological therapy', 'pale skin during the episode', 'eye deviation during the episode', 'diffuse vascular disease', 'neuropsychiatric disorders', 'presence of pacemaker', 'presence of defibrillator', 'cardio-pulmonary resuscitation', 'antihypertensive therapy', 'cardiovascular diseases', 'neurodegenerative diseases', 'peripheral neuropathy', 'immunosuppression', 'palliative care', 'situation description, like coughing, prolonged periods of straining, sudden abdominal pain, phlebotomy', 'problematic family context', 'need but absence of a caregiver', 'homelessness', 'living alone', 'chest pain', 'head or other districts trauma', 'tongue bite', 'agitation', 'foreign body in the airways', 'improvement of dyspnea', 'presence of dyspnea', 'dementia', 'general condition deterioration', 'ab ingestis pneumonia', 'further seizures in the ed', 'improvement of patient's conditions', 'neurologist consultation', 'ecg, any abnormality', 'ecg monitoring, any abnormality', 'eeg, any abnormality', 'thoracic ultrasound, any abnormalities', 'chest rx, any abnormalities', 'gastroscopy, any abnormalities', 'brain ct scan, any abnormality', 'brain mri, any abnormality', 'cardiac ultrasound, any abnormality', 'chest ct scan, any abnormality', 'pulmonary scintigraphy, any abnormality', 'abdomen ct scan, any abnormality', 'compression ultrasound (cus), any abnormality', 'performance of thoracentesis', 'administration of diuretics', 'administration of steroids', 'administration of bronchodilators', 'administration of oxygen/ventilation', 'blood transfusions', 'administration of fluids', 'heart failure', 'pneumonia', 'copd exacerbation', 'acute pulmonary edema', 'asthma exacerbation', 'respiratory failure', 'intoxication', 'covid 19', 'influenza and various infections', 'pneumothorax', 'situational syncope', 'epilepsy / epileptic seizure', 'pulmonary embolism', 'arrhythmia', 'cardiac tamponade', 'aortic dissection', 'acute coronary syndrome', 'hemorrhage', 'severe anemia', 'concussive head trauma']

Table 5: **List of the 136 Case Report Form tasks and their respective labels’ space (options)**. Several tasks share the same labels. The elements in *items* are used to define the classification tasks, and as targets s_T to quantify the relevance during the DECSELFMASK relevance attribution step.

APP: In PS condotta da 118. NOME _PERSON A alle 09.30 circa caduta a terra a dinamica non nota avvenuta al domicilio, il PARENTE non ha assistito alla caduta ma "acc" o forse dopo aver sentito il rumore. La paziente si era già alzata dal letto. Ritrovata vigile dal PARENTE. Trauma del volto contro lo spigolo di un mobile. C'è all'arrivo del 118 paziente a terra, in decubito laterale sinistro. Paziente obnubilata, respirazione spontanea o presente, GCS 13. Non apparenti deficit di lato, pupille apparentemente isocoriche, non segni di morsi usi nati il asciosfinterico. Evidenza di sanguinamento a carico della piramide nasale e FLEA livello nasale e periorbitaria dx. Somministrato acido tranexamico 1g. All'ECGRitmo da PM. Posizionato collare cervicale rigido ed effettuato immobilizzazione su spine con ragnoc. C'è AP R: C - Morbo di Alzheimer con grave deterioramento cognitivo, episodi di irrequietezza e wandering, con affaccendamento durante la notte quasi quotidiani; deambulazione instabile (visita di controllo CDCDmol in ette 6/4/2022) C - 2007 resezione di neoplasia della guancia destra; 2018 intervento di sbrigliamento C - Pregresso impianto PM bicamerale Endurity 2172 (normofunzione ante controllo del 14/05) C - Crisi epilettica recente (14/5/22) per cui avvertita terapia anti epilettica con obarbital (referita dal PARENTE adeguata compliance). TC encefalo negativa per lesioni a focolaio. EEG (2/8/22): alterazione dell'elettroencefalogramma con aspetti disfunzionali in itemporali e rari grafie elementari parossistici diffusi, inficiato da artefatti. C - Ipotroidismo C Non altri PARENTE oltre al PARENTE. NON ha tutore legale né amministratore di sostegno. C Non allergie note. C Vaccinata COVID 19 4 dosi. C Non apparentemente vaccinata con anti-tetanica. C Contatti di riferimento (PARENTE): NUM_TELEFONO

Figure 4: Impact of the DECSELFmask training on the relevance attributed by AttnLRP to the input when generating the text "Antiepileptic therapy already in place". Red represents higher relevance to be assigned to a specific token by the base model when compared to its DECSELFmask version, while green the opposite. Our model assigns much more importance to "fenobarbital", which is an antiepileptic drug, and reduces it to Alzheimer (irrelevant here). Interestingly, relevance is lowered for "Crisi epilettica" (i.e., epileptic crisis), suggesting that the model has learnt to move attention to plausible portions to the ones containing the actual information. Epileptic crisis is obviously linked to its therapy, but gives no factual knowledge about medications being in place already.

Pz che viene in ps in quanto, al termine della seduta dialitica ha presentato dispnea riferisce in oltre da circa 4/5 giorni, dolore toracico trafittivo, irradiato all'addome. Alvo scarsamente canalizzato. Eseguito tampone per SARS-CoV-2 in data 28/11 negativo. CC Compiuta vaccinazione anti SARS-CoV-2. (4 dosi) C'è AP R C'è - Ipertensione arteriosa C - Sovrappeso C - Dislipidemia C - Vasculopatia cardiaca subcritica C - IRC stadio V in dialisi, diuresi conservata - dialisi trisettimanale C - CAD. malattia coronarica trattata, PTA con sostentamento della DA media e ramo PL di CX e PTA e sostentamento su CDX prossimale (referito recente ricovero in Cardiologia per angioplastica, 3 settimane fa) C'è - BPCO in LTOT 2l/min C - Adenoma ipofisario benigno C - Sindrome ansiosa con pregressi attacchi di panico C - IPB C - Eritema migrante C TD: ASA 100mg, atorvastatina 20mg, Cmcgx2, pantoprazolo 20mg, Cmovicol, bromazepam ab, trinitrina 0'3mg colabisogni, renvela 1/2 bustina x2, trimbow 2 inalazioni x2, urorec, avodart, aerosol con 1fl. fluibron e clenil, Nobistar 1/2co, Monoket 201/2co x3, clopidogrel Disepavit 1 iniezione 2 volte/settimana (lunedì, venerdì), Parsabiv, Binocrit C NEGA ALLERGIE NOTE A FARMACI

Figure 5: Impact of the DECSELFmask training on the relevance attributed by AttnLRP to the input when generating the text "SARS-CoV-2 swab test". Red represents higher relevance to be assigned to a specific token by the base model when compared to its DECSELFmask version, while green the opposite. Our model assigns much more importance to the token "Anti SARS CoV2", and reduces it to other tokens that are indeed not required by the target.

APP: In PS condotta da 1 1 8. NOME _PERSON A alle 0 9. 3 0 circa caduta a terra a dinamica non nota avvenuta al domicilio, il PARENTE non ha assistito alla caduta ma "acc" acc o so dopo aver sentito il rumore. La paziente si era già alzata dal letto. Ritrovata vigile dal PARENTE. Trauma del volto contro lo spigolo di un mobile. "C" All'arrivo del 1 1 8 paziente a terra, in decubito laterale sinistro. Paziente obnubilitata, respirazione spontanea o presente, GCS 1 3. Non apparenti deficit di lato, pupille apparentemente isocoriche, non segni di morsure, nessun ascitoso sfinterico. Evidenza di sanguinamento a carico della piramide nasale e F LC a livello nasale e periorbitaria dx. Somministrato acido tranexamico 1 g. All'ECG ritmo da PM. Posizionato collare cervicale rigido ed effettuato immobilizzazione su spine con ragno "CC" AP R "C" - Morbo di Alzheimer con grave deterioramento cognitivo, episodi di irrequietezza e wandering, con affaccendamento durante la notte quasi quotidiani; deambulazione instabile (visita di controllo CDC D mol in ette 6 / 4 / 2 0 2 2) "C" - 2 0 0 7 resezione di neoplasia della guancia destra; 2 0 1 8 intervento di sbrigliamento "C" - Pregresso imminente PM bicamerale Endurite 2 1 7 2 (norma di un'azione ante controllo del 1 4 / 0 5) "C" - Crisi epilettica recente (1 4 / 5 / 2 2) per cui avvertita terapia antiepilettica con fenobarbital (referita dal PARENTE adeguata). TC encefalo negativa per lesioni a livello. EEG (2 / 8 / 2 2): alterazione dell'elettroencefalogramma con aspetti disfunzionali e item porali e rari grafici elementari parossistici diffusi, inficiato da artefatti. "C" - Ipotroidismo "CC" Non altri PARENTE oltre al PARENTE. NON ha tuttora legale ne'amministratore di sostegno. "CC" Non allergie note. "CC" Vaccinata COVID 1 9 4 dosi. "C" Non apparentemente vaccinata con anti-tetanica. "CC" Contatti di riferimento (PARENTE): NUM _TELEFONO

Figure 6: **Impact of the DECSELFmask training on the relevance attributed by AttnLRP to the input when generating the text "Compliance with antiepileptic therapy".** Red represents higher relevance to be assigned to a specific token by the base model when compared to its DECSELFmask version, while green the opposite. Our model assigns much more importance to the tokens "compliance", "fenobarbital", which are both relevant to the target.

Paziente ospitata di "C" Paziente accede 30 giorni fa presso PSCTO per trauma ginocchio sinistro, alla rivedenza di frattura sovracollare femore sinistro, posta indicazione a trattamento conservativo posizionale valva gessata da mantenere 30 giorni. Giunge oggi presso radiologia nostro presidio per radiodiagnostica, eseguita TC per assenza di esami precedenti, evidenzia lesione da decubito terzo medio posteriore coscia sinistra e invasi in psct: In rapporto alla scarsa compliance "I" indagine radiografica "acc" completata da acquisizione TC che evidenzia progressiva frattura intercondilare femorale distale sinistra, ingrossata, con una limitata sclerosi dei profili ossei. Presenza di protesi totale d'anca sinistra integra ed in sede. Nella norma i rapporti articolari femoro-tibiali - rotulei "CC" AP R 201 3 ESA, ipertensione arteriosa, demenza senile, pta sin, ipotroidismo "C" Paziente non deambulante da "anni"

Figure 7: **Impact of the DECSELFmask training on the relevance attributed by AttnLRP to the input when generating the text "level of autonomy".** Red represents higher relevance to be assigned to a specific token by the base model when compared to its DECSELFmask version, while green the opposite. Our model assigns much more importance to the tokens "rsa" (i.e., care home) and "Paziente non deambula da due anni" (The patient has not been able to walk for two years).